

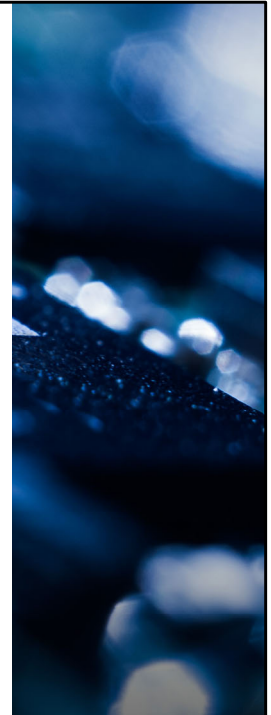


SC1006

Computer Organisation and Architecture

Computer System Overview

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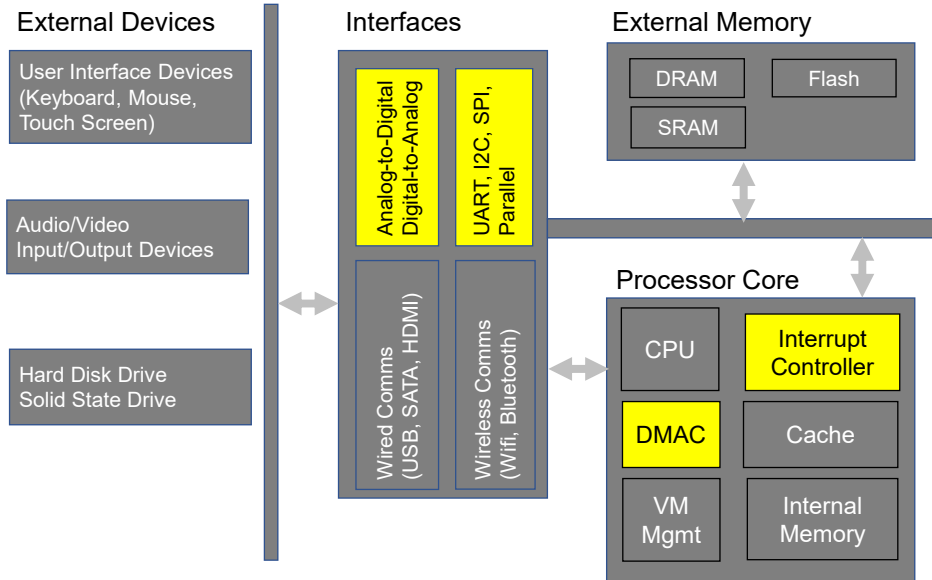


Sub-Systems in a Computer System & Presentation Schedule in Part 2

- Signal Chain Sub-System (**Week 8**)
 - ADC-DAC
 - Parallel and Serial (UART/SPI) Digital Interfaces
 - Direct Memory Access Controller
 - Interrupt Controller
- Memory Sub-System
 - Semiconductor Memories (SRAM, DRAM, FLASH)
 - Flash memory based Solid State Drives
 - Magnetic Hard Disk Drives
 - Cache Memory Management
 - Virtual Memory Management
- Communication and User Interface Sub-System (brief intro only)
 - Wired (USB, SATA, HDMI)
 - Wireless (Wifi, Bluetooth)
 - Input Devices (KBM, Capacitive Touch, Camera, Microphone)
 - Output Devices (Display, Speakers)
- More for Processor Core
 - Instruction Pipeline

we divide the computer into a few sub-systems.

Computer System Block Diagram



In this video, we will be diving deeper, into these few elements of the computer system.

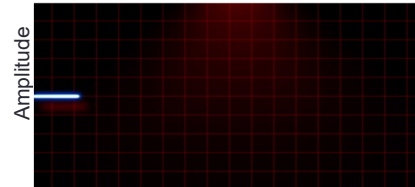
What is a Signal?

A **signal** is any **real-valued** function that changes in **time** and/or **space**.

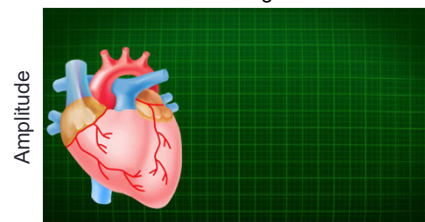
Signals that vary in time:



Waveform of audio/speech signal



Pulse signal



Electrocardiogram (ECG) signal

A signal, is any real-valued function, that changes, in time and,or space.

Cx1106 Computer Organization and Architecture

Interfacing to the Real World

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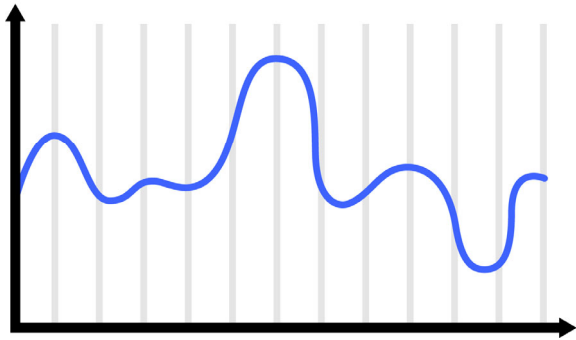
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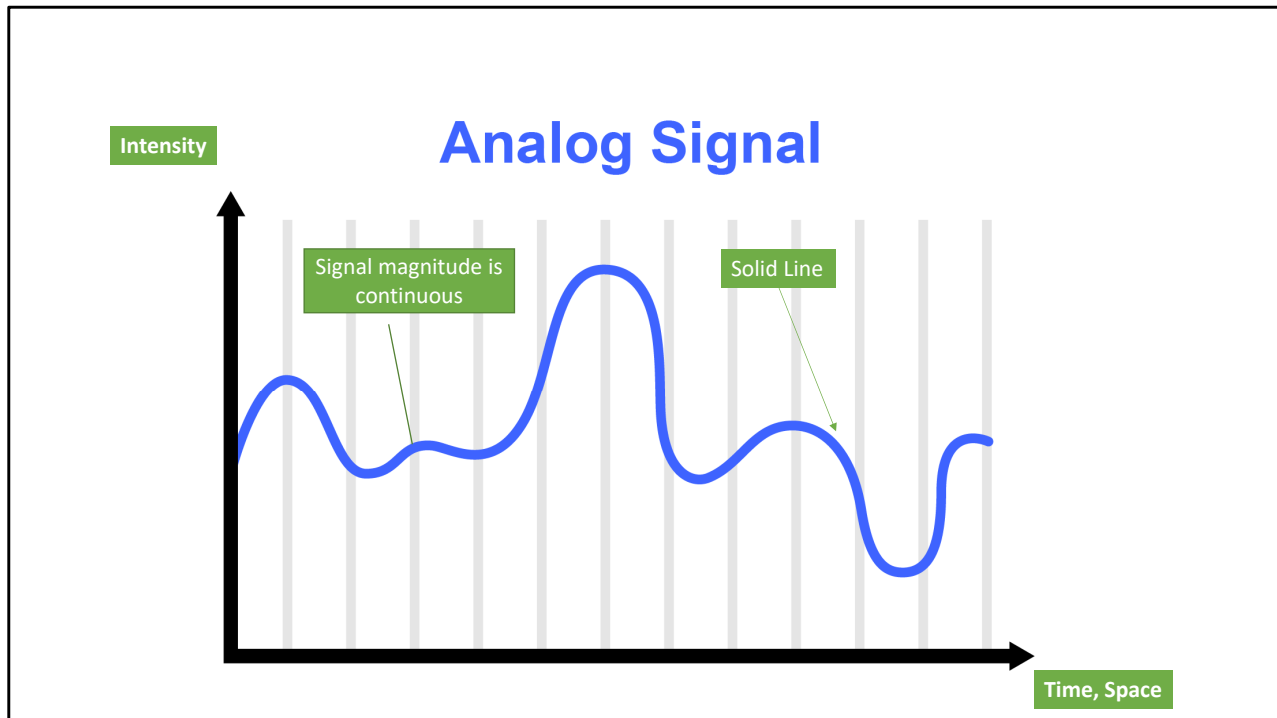
This section is about how analog real world signals are passed into the digital processor via the analog to digital converter, and how the digital output of the processor are subsequently passed back to the real world via the digital to analog converter.

Analog and Digital Signal

Analog Signal

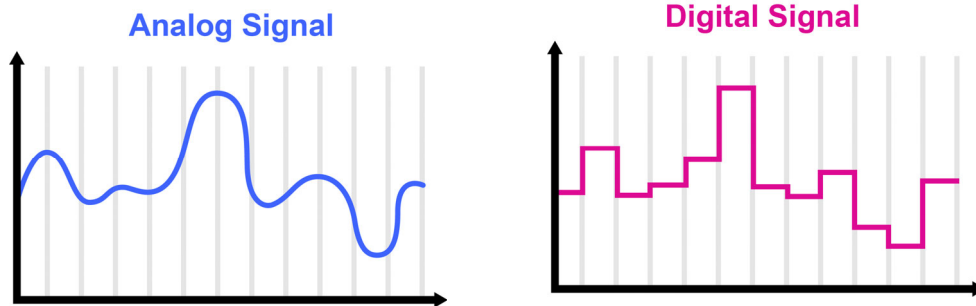


- All real world signals are actually analog, the sound we hear, the light we see, the wind we feel and the heat we sense.
- By analog, we mean that the signal magnitude is continuous. It is also continuous in time domain.



- So on a graph the intensity vs Time, an analog signal is represented by a continuous solid line.
- During the pre-digital processor days, analog signals were processed as they were in the analog domain.
 - For example, noise cancellation, audio equalization, TV brightness adjustment, audio loudness changes, etc.

Analog and Digital Signal



- Real world signals are **Analog** in nature.
 - Examples: **Sound, light, heat, pressure, smell, wind**, etc
 - In the past, most processing are done in analog domain.
 - With the introduction digital processors and decreasing cost to build them, digital processing became increasingly popular as it offer more **flexibility in implementation and are more tolerant to noise and component aging**.
- With the enhancement in semiconductor technology in the last 40 years, digital processors become sufficiently economical to be deployed in large scale and overtook analog processing as the platform of choice for signal processing.

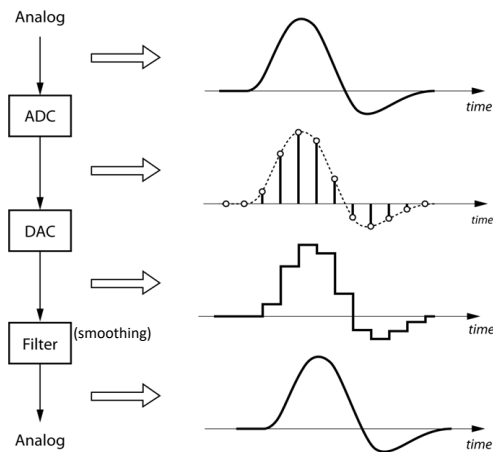
Signal Chain Sub-System



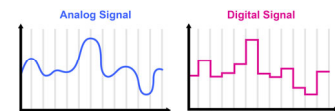
- Signal Chain Sub-System
 - ADC-DAC
 - Data transfer into and out of Processor

There are many sub-topics in the signal chain. we are going to focus on only a few of them. The A.D.C. and D.A.C. process that deals with conversion between analog and digital domain. The digital interfaces between digital modules. And the two main types of data transfer mechanism, in a computer system: the interrupt controllers and D.M.A. controllers.

Transformation between Digital and Analog Domain

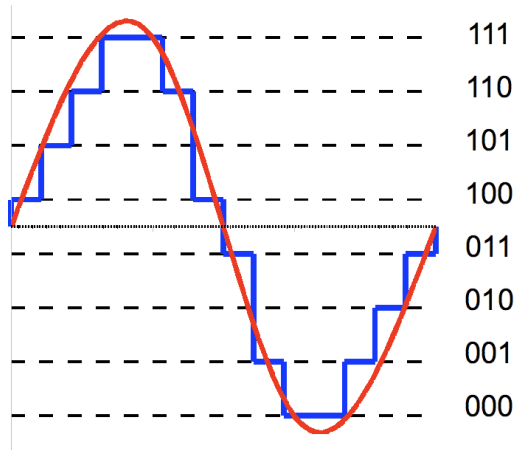


- Figure on the left shows [conversion between analog and digital domain](#).
- Digital signal is obtained by [sampling the analog signal level at discrete time](#) (known as the [sampling interval](#)).
- Sampling frequency needs to be at least twice the signal frequency ([Nyquist theorem](#)).
- Typically, the analog signal level is assigned to the [nearest discrete voltage level allowed](#) in that particular digitization process.
- Analog signal can be reconstructed back by applying a smoothing filter after DAC.

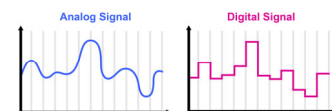


The diagram shows, the entire process of Analog-to-Digital, followed by, Digital-to-Analog conversion!!

Digital Quantization



- Figure on the left shows a typical digital quantization process of an analog signal.
- A 3-bit system is used in this example. So all signal will be mapped to one of the 8 possible representations (000 to 111).
- At each sampling point, the analog signal level is approximated to the nearest digital equivalent.

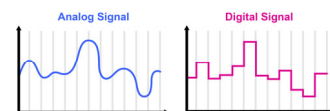


- The diagram illustrates the details of how a sine wave is digitized, using a 3-bit, A/D.C!! For a 3-bit, A/D.C!! there are 8, discrete output levels!! Level 0 to level 7!! At each sampling point, the analog signal, will be assigned, to one of the 8 values!! the one that is closest, in magnitude, to the analog magnitude will be sampled!!

Signal Chain between Processor and Real World



- Digital Processors **works only in the digital domain** so typical process these days is to convert the real-world analog signal to digital signal, allow the processor to work on the digital data, and reconstruct back the analog signal to be output to the real world.
- So **most of the interfaces we are dealing with in this course are in the digital domain.**



As mentioned earlier, digital processors, are the de facto platform, for signal processing these days!! Since a digital processor, only understands digital data... all real-world signals, would have to be digitized, before they can be processed!! Information exchange, within a computer, is typically, digital in nature, and the information will only be converted, to analog, when the signals, need to get into the real world!! So naturally, most of the interfaces, we'll be touching on in this course, is digital in nature.

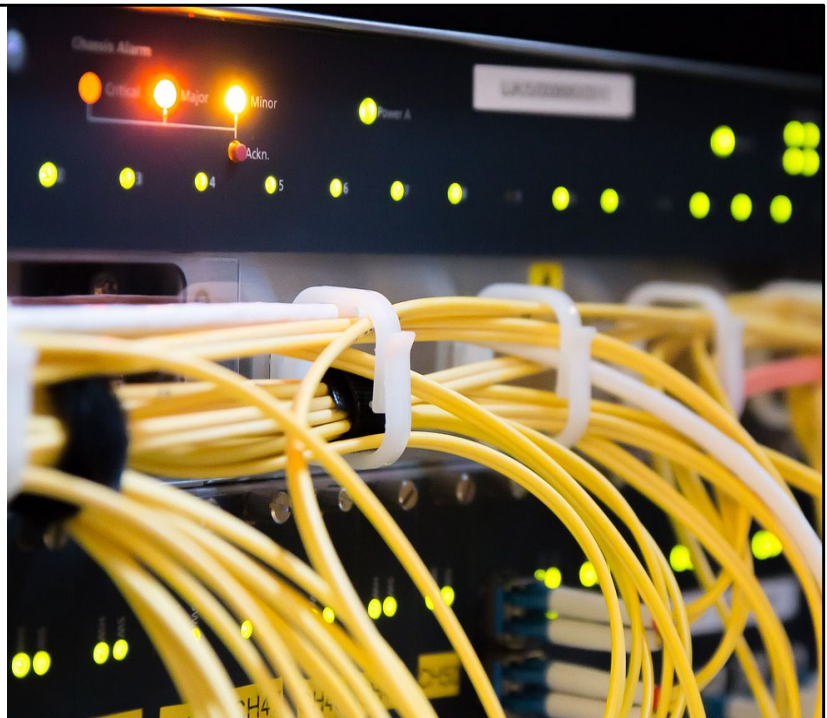
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Electrical Signal Interface

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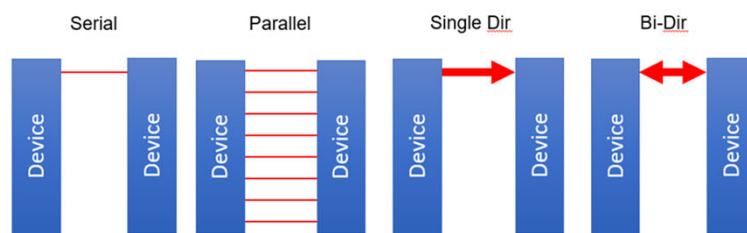
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- This section will cover the topic of electrical signal interface, where we will look at the various requirement for two devices or modules to be able to communicate with each other.

Interface

- A **boundary** where two or more devices meet to **exchange information**. Some modes of connection below
 - Single-bit Data transfer (**Serial**)
 - Multiple-bits Data transfer (**Parallel**)
- For each mode above, the connection could be **Single direction** or **Bi-direction**.



As mentioned previously, Algorithm processing these days is predominantly done in the digital domain.

- this is why most of the interface we are going to discuss in this course are digital interfaces.
- Now, before that, what exactly is an interface?
 - In computer terms, interface is a boundary where two or more devices meet to exchange data.
- There are different ways in which devices can be connected.

Interface Compatibility

- Interfacing one electronic device to another requires compatibility in
 - **Electrical signal level** (to be explained in the slides next)
 - **Communication protocol** (Handshaking and Data signals; to be explained in the next two sub-sections).



- This is a very simple slide, but, it delivers, two pieces of important information!!for two devices to talk, two criteria have to be met!!First, the electrical signal level, of the two devices ,has to be compatible!!Second, the devices, must be speaking, in the same language. For our case, they must use, the same communication protocol!!We will discuss, the two points, in more details in later slides!!

Electrical Signal level (Safety)

- Primary consideration when connecting two electrical device together.
- Ensure that the output voltage level of output device do not exceed the maximum allowable input voltage level of the input device.
- Electronic devices will either get 'fried' i.e. spoilt or have its reliability reduced if the input voltage is higher than what they are designed for.
- So do check the voltage level which the device input/output is operating on (1.8V, 3V, 5V, 15V etc) before connecting them together.



- There are two aspects ,where the Electrical Signal Level, is concerned!!First! is the safety consideration!!When connecting, two electrical devices, the most important thing, you need to remember, is safety!!We are not, dealing with high power electronics here, but, if you made the wrong connection, you may still see some smokes, coming out from your equipments and boards!!So, Make sure, the output voltage, and input voltage, of the two devices, are compatible. For example, don't connect, a 5 Volt device, to a 3 Volt device, unless you know what you are doing!!Electronics devices, in general, do not like to be, electrically stressed. So. it'll either get fried, also known as spoilt, or its reliability, will be reduced, if a voltage higher, than what it is designed for, is applied to the device!!So do check out, the input output voltage rating, of each of the devices, you are connecting!!!

Electrical Signal Level (Digital Data Transfer)

- A voltage as input: Logic '1' or '0' ?
- Four parameters
 - **VOH.**
 - Transmitting a Logic '1' yield a signal level of $\geq$ VOH
 - **VOL.**
 - Transmitting a Logic '0' yield a signal level of $\leq$ VOL
 - **VIH.**
 - A received signal with level $\geq$ VIH will be recognized as a Logic '1'
 - **VIL.**
 - A received signal with level $\leq$ VIL will be recognized as a Logic '0'
- These are for digital logic level compatibility.



- When a device receive a signal at its input pin, is it considered Logic '1' and '0'?
- It depends on a few device parameters shown here
- These four parameters dictate whether two devices are able to communicate with each other properly or not.
- Their definition are as follows
 - VOH determines the min voltage that a device will transmit if it is transmitting a logic '1'
 - Similarly, if a device transmit a logic '0', it will transmit a signal with magnitude no larger than VOL.
 - On the receive side, we have VIH and VIL.
 - For a signal to be recognised as a logic '1', its has to at least VIH.
 - For a signal to be Logic '0', its electrical level has to be less than VIL.

Electrical Signal Level (Digital Data Transfer)



- **Device #1**
 - Device #1 datasheets: $V_{OH} = 2.4V$, $V_{OL} = 0.4V$
 - Output Voltage level for logic '1' = $V1+ \geq V_{OH} = 2.4V$
 - Output Voltage level for logic '0' = $V1- \leq V_{OL} = 0.4V$
- **Device #2**
 - Device #2 datasheets: $V_{IH} = 2.0V$, $V_{IL} = 0.8V$

Device #2 senses the logic level correctly, because

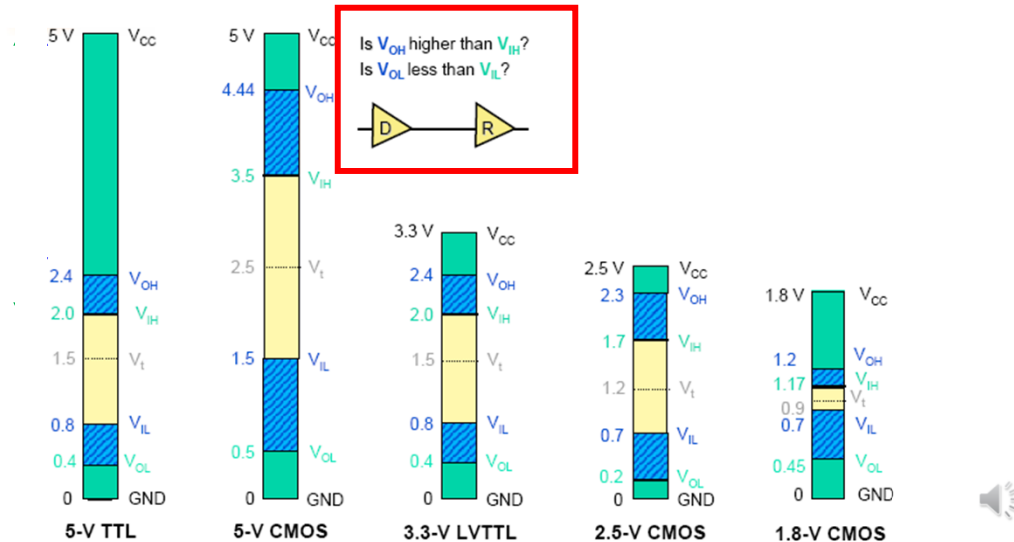
$$V1+ \geq 2.4V > V_{IH} = 2.0V$$

$$V1- \leq 0.4V < V_{IL} = 0.8V$$



- This slide further illustrate what we discussed in the previous slides. Device number 1 is the transmitter, $V1+$, correspond to Logic 1, and $V1-$ minus, correspond to Logic 0!!! if we assume, that $V_{OH} = 2.4$ Volt, and $V_{OL} = 0.4$ Volt, for Device 1, then $V1+$, will be greater, or equal to, 2 point 4 Volt. while $V1-$ minus, will be smaller or equal to 0 point 4 Volt. These parameters, are design dependent ,and are specified, in the device datasheets. You may have studied this, in your digital, logic course as well. When the signal ,reaches Device 2, If we assume that $V_{IH} = 2$ Volt, and $V_{IL} = 0.8$ Volt, the signal ,will be recognized, as a logic 1 correctly, since its signal level, is larger than 2 Volt!! It will be, recognized as a logic 0 correctly, since its signal level, is less than 0 point 8 Volt!! These parameters, can be found ,in the device datasheets. In summary, two devices, will be able to talk ,to each other correctly, since their V_{OH} , V_{OL} , V_{IH} , and V_{IL} are compatible, in this case study!!

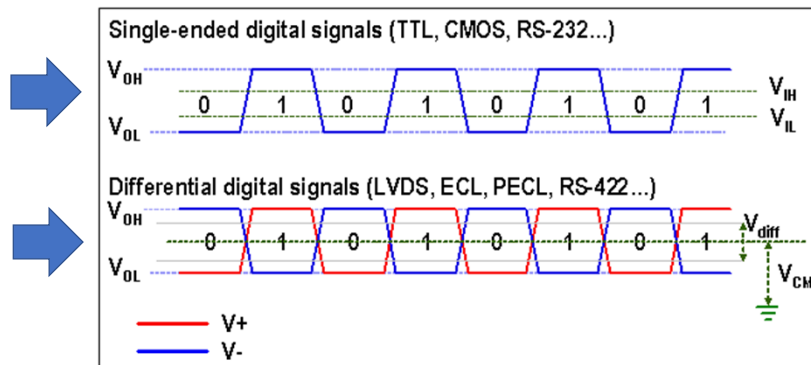
Electrical Signal Level Standards



- Depending on the voltage rating, and the chip design, the four parameters V_{OH} , V_{OL} , V_{IH} and V_{IL} can be quite different. This chart shows, some of the devices, out there. It is not, the complete list, and there are many others. The Bottom line is, please check, the device datasheets, before you decide, to hook them up together. You can always ask, two key questions, on the slide. V_{OH} , highlighted by blue line, needs to be higher, than V_{IH} , highlighted by green line. V_{OL} needs to be lower than V_{IL} always. These are to cater for noise, or losses, as the electrical signal, propagate thru, the circuits.

Differential Signals

- Differential signals has better noise tolerance so is able to be clocked at higher frequency.
- V_{CM} = Common Mode Ground.

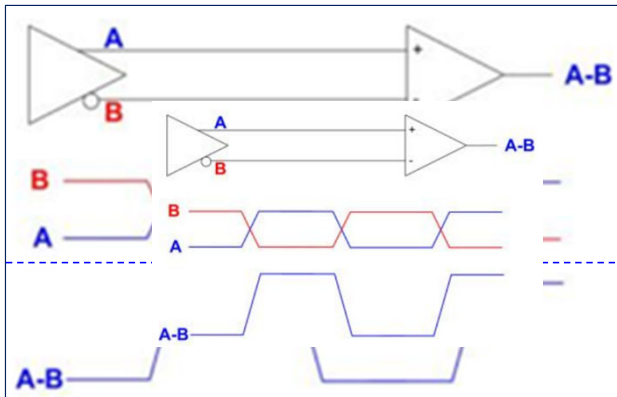


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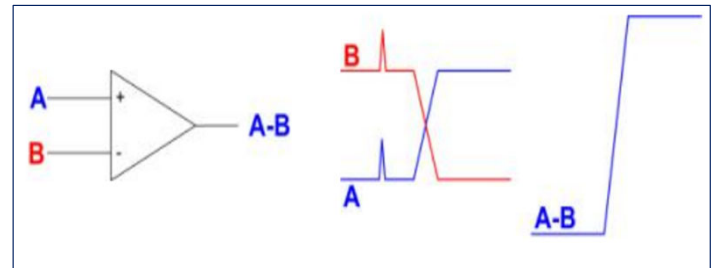
- (Differential signaling and crosstalk => <https://resources.system-analysis.cadence.com/blog/msa2021-the-advantages-of-differential-signaling>)
- What we have seen, in the previous slides, are single ended signal. ,Electrical signals, can also be transmitted, via differential signals,. V_+ , is the original signal ,while V_- , is its inverted signal.,If V_+ , is greater than V_- , then ,it is a logic 1,. if V_- , is greater than, V_+ , then it will be, a logic 0., More explanation, is to be presented, in the next slide, for Differential signaling.

Differential Signals

Differential signals have better noise tolerance



A Larger Margin



External interference canceled

A denotes V+

B denotes V-

Differential signals has better noise tolerance in general

- Because they have a larger margin. You can see in the illustration on the left that the voltage for "1" and "0" are further apart and can accommodate more noise.
- You can also see in the illustration on the right that Any external interference will also have the same effect on A and B, so the difference between them remains unchanged (due to cancellation).